Conditional Generative Adversarial Network-Based Training Sample Set Improvement Model for the Semantic Segmentation of High-Resolution Remote Sensing Images

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***Abstract*— To achieve high segmentation quality, deep semantic segmentation neural networks (DSSNNs) need to be trained on diverse direction, location, and neighboring category combinations for each pixel in the input–output image patches. Achieving this goal requires a large sample set. However, in many practical application scenarios, a very large training sample set is too expensive to achieve, or insufficient remote sensing image data are available. These limitations directly affect the quality of the results of DSSNNs. To address the above-mentioned problem, this article proposes a conditional generative adversarial network (CGAN)-based training sample set improvement model (CGAN-TSIM) for the semantic segmentation of high-resolution remote sensing images. In CGAN-TSIM, the generator model of the CGAN can generate a sample image when a ground-truth image is an input as a "condition." A condition generation mechanism is designed to create ground-truth images, and these ground-truth conditions are used to drive the CGAN to generate samples containing more diverse object combinations, directions, and locations. These generated images can be added to the original training sample set to improve their spatial information diversity. Rather than simply relying on passively finding samples that contain diverse spatial information, CGAN-TSIM extracts high-level spatial information from the original training images and actively generates new sample images. Experiments show that the samples generated by CGAN-TSIM can improve the quality of the sample set. Compared with other traditional methods, CGAN-TSIM enables better classification accuracy.**

***Index Terms*— Condition generation, generative adversarial network (GAN), high-resolution remote sensing classification, sample generation, semantic segmentation.**

I. Introduction

As HIGH-RESOLUTION remote sensing satellites, airplanes, and unmanned aerial vehicles have increasingly been adopted for image acquisition, large volumes of high-resolution images have become available [1], [2]. Semantic segmentation methods, which divide an image into several regions with certain category labels, are important in the end-to-end automatic processing of such remote sensing images [3], [4]. The traditional shallow methods used for segmentation, such as pixel threshold, level set, clustering, and graph methods, are based on low-level features and have difficulty achieving acceptable results due to the increased information complexity in high-resolution remote sensing images [5]. As an alternative, deep learning methods, which can extract higher level features to improve classification accuracy, have gradually gained prominence in the field of high-resolution remote sensing image semantic segmentation [6], [7].

A deep semantic segmentation neural network (DSSNN) is a typical deep learning model that outputs semantic segmentation result corresponding to an image patch [8]. Currently, fully convolutional networks (FCNs), SegNet, and U-Net, are the mainstream base models used in DSSNNs; many methods have improved upon these three basic models to achieve specific application objectives in the remote sensing field. FCNs include deconvolution layers to realize upsampling and dense pixel output [9]; SegNet and U-Net include encoding and decoding structures that enforce the transformation of spatial information from input to output [10], [11]. DSSNNs can perform end-to-end parsing or classification of remote sensing images and achieve a high degree of automation and accuracy [12]–[15]. Through the introduction of additional network structures or training processes, FCNs can be adapted for application to airborne light detection and ranging (LiDAR), synthetic aperture radar, and digital terrain model data to achieve improved accuracy [16]–[18]. The intermediate features of an FCN and the uncertainty of the output pixels can also be taken as additional information to improve the segmentation of ground objects [19], [20]. Inspired by the encoding and decoding structures of SegNet, multimodal and multiscale remote sensing image segmentation can be achieved [21]. Considering contextual spatial relations can also help improve the segmentation results of FCNs; by more effectively extracting border, structural or contextual information, DSSNNs allow better detection results to be obtained for buildings, roads, and ground objects [22]–[25]. Moreover, the encoder and decoder of U-Net can be reconstructed to improve object boundaries and small features [26]. Through

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a combination of residual connections, atrous convolutions, and additional pooling masks, the remote sensing segmentation accuracy of U-Net can be improved [27].

A DSSNN is a data-dependent model. To achieve high segmentation quality, a DSSNN needs to be trained on diverse combinations of directional, locational, and neighbor information for each pixel in the input–output image patches [28]. A lack of spatial diversity in the training data set and the loss of spatial information in convolution/deconvolution operations can lead to defects in the object boundaries and shapes in the DSSNN output [9]–[11], [29], [30]. In the field of computer vision, many very large general-purpose data sets have been constructed (e.g., the Pascal and COCO data sets) that can assist in efficient semantic segmentation. However, it is difficult to construct similarly large sample data sets of remote sensing images for segmentation for the following reasons.

A. Content of Remote Sensing Data Has Regional Characteristics

In contrast to general objects, such as cats, bicycles, and humans, which are the typical targets in data sets for computer vision research, the objects in remote sensing images will show variations in color, texture, and size among different regions even when those objects belong to the same category. For example, houses and vegetation in tropical areas are significantly different from those in temperate areas; thus, samples from one region may not help to improve the segmentation ability of a model for other regions. As a result, when the research area is restricted to a certain limited region, the number of suitable/available remote sensing images will decrease.

B. Even Within a Single Remote Sensing Image, the Category Structures Considered for Different Applications Are Different

For example, the categories considered in a remote sensing image for an urban application may include "buildings" and "vegetation," whereas, for an agricultural application, the category "vegetation" is too general and needs to be further subdivided into different crop categories. This characteristic makes the establishment of a general category structure infeasible for practical remote sensing applications, and the need for a specific category structure further restricts sample acquisition.

For the above-mentioned reasons, obtaining a sufficiently large training sample set of remote sensing images for semantic segmentation is both difficult and costly. The resulting lack of training data limits the segmentation quality of DSSNNs in remote sensing applications.

For classification tasks with insufficient samples and large feature space, semisupervised methods are a typical solution; by utilizing posterior class distributions of remote sensing data, the classification accuracy for object or segment recognition can be greatly improved [31]–[33]. For deep neural networks, whose performance strongly depends on the training data, semisupervised training goals can also be achieved by adopting generative adversarial networks (GANs) [34]. A GAN model consists of two modules: a generator (G), which generates samples, and a discriminator (D), which estimates the probability that a given sample either comes from the real training set or was generated by G [35]. Through adversarial training (AT) between G and D, the generation ability of G and the recognition ability of D will simultaneously improve. In the field of remote sensing, the main goals when training a GAN are to enhance the recognition ability of D and prevent overfitting. Through AT, crucial features of hyperspectral images can be revealed, and the classification ability of D can be enhanced [36]–[38]. GANs can further improve the classification ability for small objects, boundaries and other details [39], [40]. For scene classification, GANs can also improve the classification accuracy or can be used to extract more useful features from scene images [41]–[43].

The output of the discriminator in a GAN is a decision regarding whether the input image is "real or fake." Because of the binary nature of this decision, it is not possible to design a discriminator as a semantic segmentation neural network; therefore, no traditional "semantic segmentation GANs" exist. An alternative way to use a GAN to improve the classification ability for a task is to use the generator to generate sample data and then use these data to train another classifier model [44]. For scene classification, generating samples in this way can improve the sample set diversity and lead to higher classification accuracy when the number of real training samples is small [45], [46]. This concept can also be applied for semantic segmentation by generating more diverse samples to enhance the classification ability of a DSSNN model. However, a training sample for semantic segmentation consists of both an image patch and a corresponding pixelwise segmentation image. Therefore, the traditional GAN structure, in which the input consists of a "random latent variable + category label" and the output is a generated image, cannot be used to create samples suitable for semantic segmentation purposes.

A conditional GAN (CGAN) is an image-to-image translation model [47]; a CGAN can learn the relationship between input images and output images and then perform style or domain transfer accordingly [48]. A CGAN-based method can be used to extract the important features among the 2-D images of a scene to address the camera localization problem [49]. For laser scanning point cloud data, a CGAN can be used to optimize the structure completion, extrusion removal, and regularization processes for the corresponding 3-D model [50]. Using a CGAN, we can achieve the goal of inputting a ground-truth image patch and generating a fake remote sensing image patch that is close to a real image. Therefore, this article proposes a CGAN-based training sample set improvement model (CGAN-TSIM) for the semantic segmentation of high-resolution remote sensing images. In CGAN-TSIM, through AT, the generator of the CGAN is trained to generate a remote sensing image from an input ground-truth image serving as a "condition." A condition generation mechanism is designed to create ground-truth images, and these ground-truth conditions are used to drive the CGAN to generate samples containing more diverse object combinations, directions and locations. These generated images can be added to the original training sample set to improve their spatial information diversity.

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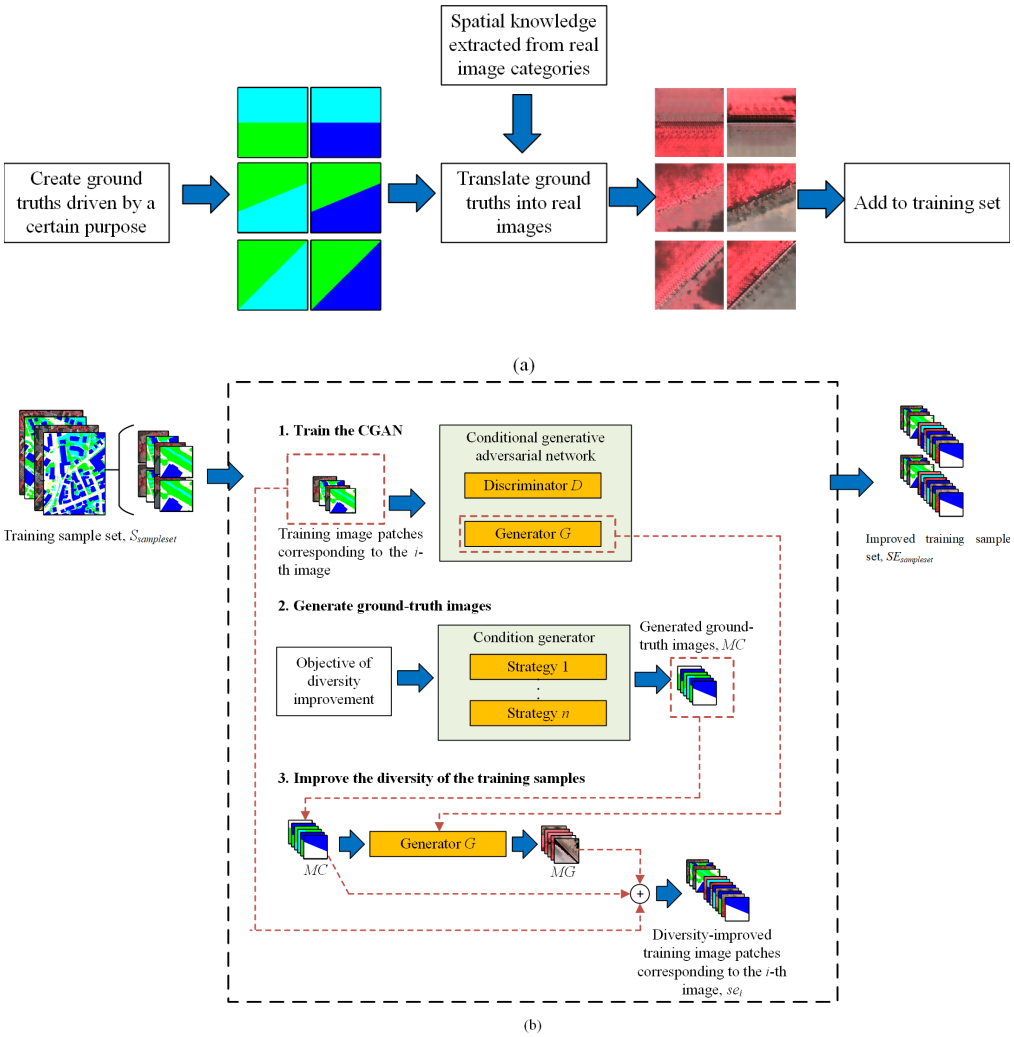


Fig. 1. Concept and process of the proposed method. (a) Concept of active sample generation. (b) Overall process of the proposed method.

Rather than relying on passively finding samples that contain diverse spatial information, CGAN-TSIM can extract high-level spatial information from the original training images and actively generate additional appropriate sample images. Experiments show that the samples generated by CGAN-TSIM can realistically represent the spatial information of and relationships among ground objects, and these generated samples can improve the quality of the sample set, preventing the classification model from becoming overfitted to the particular details of the original training images. Compared with other traditional sample set enlargement methods or AT enforcement methods, CGAN-TSIM enables better classification accuracy, especially for the boundaries of ground objects.

II. Methodology

A. Overall Process of the Proposed Method

For good semantic segmentation results to be obtained, the training set needs to contain sufficient diversity in terms of object directions, locations, and neighboring category combinations to train or verify each pixel output of the DSSNN. To meet this requirement, the training set must contain a representative and comprehensive set of condition possibilities, and thus, must include an enormous number of remote sensing images. In most practical applications, this requirement is too expensive to achieve, or insufficient image data are available. To address this problem, the main idea advanced in this study is that instead of struggling to find sufficiently diverse image patches in the original images, it is more efficient to generate the required samples directly. The concept of the proposed method is shown in Fig. 1(a).

As shown in Fig. 1(a), to enhance the spatial information diversity of the sample set, we can create ground-truth images containing different object directions, locations, and neighbor combinations and then transform these ground-truth images into remote sensing images, supported by spatial knowledge extracted from real-image categories. In this way, directions and combinations that do not exist in the original images can be easily and autonomously generated. Then, these generated images can be added to the training set to improve their diversity of spatial information.

Based on the concept illustrated in Fig. 1(a), this article proposes a CGAN-based CGAN-TSIM for the semantic segmentation of high-resolution remote sensing images. The overall process of CGAN-TSIM is shown in Fig. 1(b). Initially, there are N_{img} remote sensing images and corresponding ground-truth images. Each pair consisting of an image and its

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ground truth is cut into training image patches. These training image patches form the original training sample set $S_{\text{sampleset}} = \{s_1, s_2, \ldots, s_{\text{Nimg}}\}$, where s_i is the group of N_{ori} training image patches corresponding to the ith remote sensing image, $s_i = \{(\text{mr}_1, \text{mt}_1), (\text{mr}_2, \text{mt}_2), \ldots, (\text{mr}_{\text{Nori}}, \text{mt}_{\text{Nori}})\}$, with mr_j being the jth patch of the remote sensing image and mt_j being the corresponding ground truth for mr_j. The improvement process for each s_i is divided into the following three steps.

1) Train the CGAN: The samples in s_i are used to train a CGAN. Through AT between the generator (G) and the discriminator (D), spatial knowledge associated with each category is extracted from s_i. G can be given a ground-truth image as a "condition;" then, based on this condition, G translates the spatial information into specific locations in a generated remote sensing image.
2) Generate ground-truth images: Using a condition generator (CG) model that includes algorithms for multiple ground-truth generation strategies, the objective of improving the spatial information diversity of the images can be represented in the form of parameters for the strategy algorithms. By providing suitable input parameters to the CG, we generate a set of N_{cnd} ground-truth images, $\text{MC} = \{\text{mc}_1, \text{mc}_2, \ldots, \text{mc}_{\text{Ncnd}}\}$.
3) Improve the diversity of the training samples: The generator G in the CGAN takes MC as the input conditions and generates a set of corresponding remote sensing images, $\text{MG} = \{\text{mg}_1, \text{mg}_2, \ldots, \text{mg}_{\text{Ncnd}}\}$. Then, MC and MG are combined to form a new training set $s_{\text{new}-i} = \{(\text{mg}_1, \text{mc}_1), (\text{mg}_2, \text{mc}_2), \ldots, (\text{mg}_{\text{Ncnd}}, \text{mc}_{\text{Ncnd}})\}$. Here, $s_{\text{new}-i}$ is a generated sample set based on the spatial diversity improvement objective adopted in step (2), and in $s_{\text{new}-i}$, the spatial information contained in the ith training image is manifested in a controllable manner. The set $s_{\text{new}-i}$ and the set s_i are combined to form a new set, $\text{se}_i = s_{\text{new}-i} \cup s_i$, where se_i consists of the improved training samples corresponding to the ith remote sensing image.

The above-mentioned three steps are repeated for each original remote sensing image to obtain an improved training sample set $\text{SE} = \{\text{se}_1, \text{se}_2, \ldots, \text{se}_{\text{Nimg}}\}$. Based on the above-mentioned process, the CGAN-TSIM algorithm is presented as pseudo-algorithm 1.

Pseudo-Algorithm 1 CGAN-TSIM Algorithm

Input: $S_{\text{sampleset}}$
Output: SE
Begin
SE = ø;
CGAN = Create a CGAN model;
CG = Assign strategies to the CG;
for s_i in $S_{\text{sampleset}}$
  Use s_i to train CGAN;
  MC = ø; MG = ø; $s_{\text{new}-i}$ = ø;
  for sg_i in CG's strategies
    MC ← (sg_i-created ground-truth images);
  for mc_i in MC
    MG ← Input mc_i into the CGAN's generator G and obtain the generated results;
  for (mg_i, mc_i) in (MG, MC)
    $s_{\text{new}-i}$ ←(mg_i, mc_i);
  $\text{se}_i = s_{\text{new}-i} \cup s_i$;
  SE ← se_i;
return SE;
END

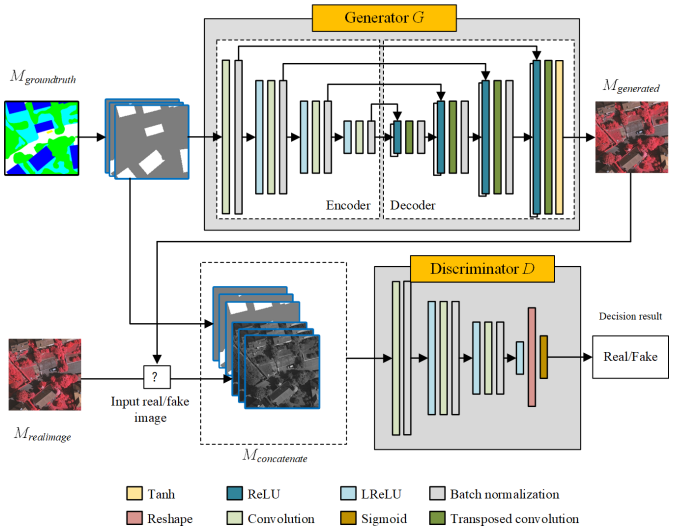


Fig. 2. CGAN architecture adopted in CGAN-TSIM.

B. Conditional Generative Adversarial Network for Remote Sensing Image Patch Generation

CGAN-TSIM uses a CGAN to generate remote sensing image patches. The CGAN architecture adopted in this research is shown in Fig. 2.

As seen in Fig. 2, the CGAN consists of two modules: a generator (G) and a discriminator (D). The generator G is further divided into two parts, namely, an encoder and a decoder, each of which is composed of multiple stacks of layers. The encoder layer stacks consist of leaky rectified linear unit (LReLU) layers and convolutional and batch normalization (BN) layers (the first stack does not include an LReLU layer). The decoder layer stacks consist of ReLU, transposed convolutional and BN layers, and a layer with the tanh activation function is finally used as the output layer. The kernel size for all of the convolutional and transposed convolutional layers is 5×5, and the stride is 2×2. With these settings, the size of the input feature map is reduced by half after each encoder layer stack and is doubled after each decoder layer stack. To improve the pixel-level generation accuracy, G includes skipping connections between encoder and decoder layer stacks of the same size. For the discriminator D, which has the same stack structure as the encoder of G, reshape and sigmoid layers are added after the last stack to produce a decision.

The input to G is a ground-truth image patch that serves as a condition, $M_{\text{groundtruth}} = \{m_1, m_2, \ldots, m_{N\text{groundtruth}}\}$, where $N_{\text{groundtruth}}$ is the number of bands in the ground-truth image and m_i is the ith band of $M_{\text{groundtruth}}$. Through the encoder and decoder process, G creates a fake image $M_{\text{generated}}$,

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which may be highly similar to the real image $M_{\text{realimage}}$. Both $M_{\text{realimage}}$ and $M_{\text{generated}}$ consist of N_{bands} image bands $\{I_1, I_2, \ldots, I_{\text{Nbands}}\}$, and each band is normalized to the range $[-1, 1]$. The input to D is $M_{\text{concatenate}}$, which is a concatenation of the bands of either $(M_{\text{groundtruth}}, M_{\text{realimage}})$ or $(M_{\text{groundtruth}}, M_{\text{generated}})$; therefore, the number of bands of $M_{\text{concatenate}}$ is $N_{\text{bands}} + N_{\text{category}}$. The output of D is a decision regarding whether the input was formed from a real or fake image. The objective of G and D can be expressed as follows:

$$L_{\text{CGAN}}(G, D) = E[\log D(M_{\text{groundtruth}}, M_{\text{realimage}})] + E[\log(1 - D(M_{\text{groundtruth}}, G(M_{\text{groundtruth}})))]. \tag{1}$$

For the pixel-level generated image, the l-one-norm distance between the generated image and the real image also needs to be considered

$$L_{\text{distance}}(G) = E[l1\text{norm}(M_{\text{realimage}} - G(M_{\text{groundtruth}}))]. \tag{2}$$

Therefore, the overall CGAN objective is as follows:

$$\arg \min_G \max_D L_{\text{CGAN}}(G, D) + \lambda L_{\text{distance}}(G). \tag{3}$$

where λ is a distance weight. The goal of training the CGAN is achieved by optimizing (3) [47]. Through training, the images generated by G gradually become more similar to real remote sensing images. Thus, G can be used as a sample generation component in CGAN-TSIM to obtain new samples.

C. Condition Generator

In CGAN-TSIM, a CG is used to create new ground-truth images. The objective of the CG is to create ground-truth images that contain a greater variety of object direction, size, and category combinations than the original images. Based on these ground-truth images, the CG drives G to produce corresponding "real" remote sensing images to improve the sample set diversity.

The output of the CG is a set of N_{cnd} created images, $\text{MC} = \{\text{mc}_1, \text{mc}_2, \ldots, \text{mc}_{\text{Ncnd}}\}$, where mc_i is an N_{category}-color image, with N_{category} being the number of categories in the ground-truth image, whose corresponding colors are listed in Colorplate $= \{\text{cp}_1, \text{cp}_2, \ldots, \text{cp}_{\text{Ncategory}}\}$. To process different types of ground objects, the CG can adopt a variety of strategies for creating ground-truth images. In this study, we design two strategies: an area separation strategy and an object placement strategy. The processes of these two strategies are shown in Fig. 3.

1) Area Separation Strategy: Ground objects such as roads and buildings occupy a certain area, but their sizes and shapes are diverse; therefore, these types of objects need to be represented in as many combinations and directions as possible in the training set. For the area separation strategy, as shown in Fig. 3(a), we consider a square SQ whose width is W. SQ is separated by a line

$$\text{line}(x, y, \theta, d) = \begin{cases} y - \tan(\theta)(x + d), & \theta \neq \dfrac{\pi}{2} \\ x, & \theta = \dfrac{\pi}{2}. \end{cases} \tag{4}$$

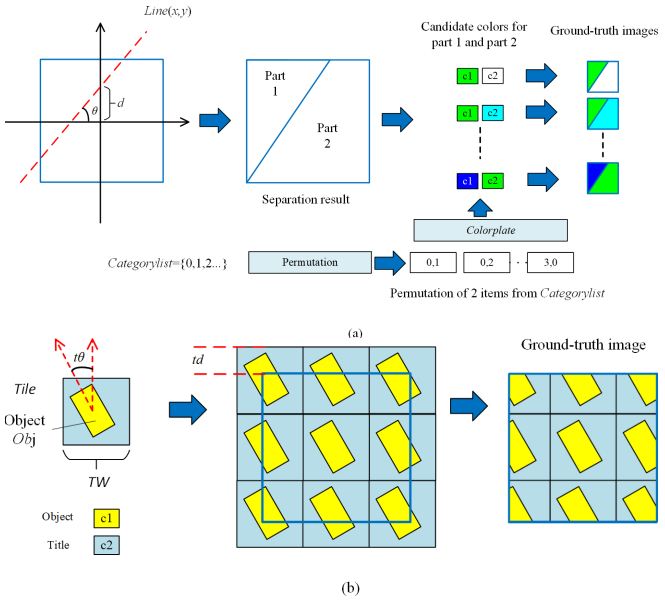


Fig. 3. Processes of the two ground-truth image creation strategies. (a) Area separation strategy. (b) Object placement strategy.

Here, $\text{line}(x, y, \theta, d)$ is determined by the slope angle θ and the intercept d. The line equation separates SQ into two parts based on θ and d, which determine the direction and proportions of this separation. Thus, varying θ and d results in multiple diverse separations. To create a ground-truth image, the two separated parts of SQ then need to be filled with corresponding category colors. The candidate categories to which SQ can be assigned are listed in Categorylist $= \{c_1, c_2, \ldots, c_n\}$. A candidate category label filling strategy for SQ can be generated by permuting two items from Categorylist. The category labels assigned to SQ are then translated into colors based on Colorplate. The algorithm for the area separation strategy is given as pseudo-algorithm 2.

In the area separation strategy algorithm, the input parameters are $P_{\text{angleincrement}}$ and P_{offset}. $P_{\text{angleincrement}}$ controls the stride with which the slope angle θ varies from 0 to π, and P_{offset} controls the stride with which the intercept d varies from $-W/2$ to $W/2$. Based on different combinations of $P_{\text{angleincrement}}$ and P_{offset}, diverse directions and sizes of the two parts of SQ can be achieved. Thus, the area separation strategy algorithm creates ground-truth images and adds them to MC.

2) Object Placement Strategy: Certain objects (e.g., cars) have relatively fixed shapes; the object placement strategy addresses this situation. As shown in Fig. 3(b), the input is an image tile (Tile) of width TW that contains an object (obj). The shape and size of obj are determined by the object's category, and its direction is $t\theta$. Multiple instances of Tile can be combined to form a larger image. A frame SQ is then used to crop an image patch from this larger image. The offset of the cropping position is td. In this strategy, the object size and shape are fixed, but changing $t\theta$ and td will result in different subimages, producing a variety of ground-truth images. The algorithm for the object placement strategy is given as pseudo-algorithm 3.

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Pseudo-Algorithm 2 Area Separation Strategy Algorithm

Input: $P_{\text{angleincrement}}$, P_{offset}
Output: MC
Begin
$\theta = 0$; pl = ø; cl = ø; zl = ø; MC = ø;
while ($\theta < \pi$)
  d =-W/2;
  while ($d < W/2$)
    pl $\leftarrow$ (θ, d); $d = d + P_{\text{offset}}$;
  $\theta = \theta + P_{\text{angleincrement}}$;
for ($p1$, $p2$) in Permutations(Categorylist, 2)
  $c1$ = color vector of $p1$ in Colorplate;
  $c2$ = color vector of $p2$ in Colorplate;
  cl $\leftarrow$ ($c1$, $c2$);
for row in range(1, W)
  for col in range(1, W)
  x = col-W/2; y = -(row-W/2);
  zl $\leftarrow$ (row, col, x, y);
for (θ, d) in pl
  for ($c1$, $c2$) in cl
    M = Create an image whose width is W;
    for (row, col, x, y) in zl
      v = input x, y, θ, and d into Eq. (4);
      if $v >= 0$ then M [row, col] = c1;
      else M [row, col] = c2;
    MC $\leftarrow$ M;
return MC;
END

In the object placement strategy algorithm, the input parameters are $P_{\text{objectangle}}$ and $P_{\text{tileoffset}}$. $P_{\text{objectangle}}$ controls the stride with which the object direction angle $t\theta$ varies from 0 to π, and P_{offset} controls the stride with which the offset td varies from 0 to TW. These two parameters lead to diverse cropping results and the creation of ground-truth images with rich spatial diversity.

III. Experiments and Results

A. Experimental Data

In this research, we constructed two test data sets based on the data set for the "semantic labeling contest of the International Society for Photogrammetry and Remote Sensing (ISPRS) Working Groups (WG) III/4 data set" as experimental targets. The details of these data sets are as follows.

1) Vaihingen Data Set: We selected five images from the original Vaihingen data set. Each image contains three bands (near-infrared, red, and green), and the resolution is 9 cm.

2) Potsdam Data Set: We selected five images from the original Potsdam data set. Each image contains three bands (red, green, and blue), and the resolution is 5 cm.

There are six categories in the data set for the "semantic labeling contest of the ISPRS WG III/4 data set:" impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Because very few areas in the Vaihingen data set belong to the clutter/background category, we removed this category from this data set. The images and ground truths from the two data sets are shown in Fig. 4.

Pseudo-Algorithm 3 Object Placement Strategy Algorithm

Input: $P_{objectangle}$, $P_{tileoffset}$
Output: MC
Begin
pl = ø; MC = ø; $t\theta = 0$; td = 0;
$c1$ = Object color in Colorplate;
c2 = Background color in Colorplate;
N_{tiles} = ceil(W/TW)+1;
while ($t\theta < \pi$)
  td = 0;
  M_{Tile} = Create a TW × TW image whose color is $c2$;
  Draw an object whose direction is $t\theta$ and whose color is $c1$;
  M_{big} = Combine $N_{\text{tiles}} \times N_{\text{tiles}} M_{\text{Tile}}$ into a larger image;
  while (td < TW)
    M = Crop M_{big} at [td: td+W, td: td+W];
    MC $\leftarrow$ M;
    td = td + $P_{\text{tileoffset}}$;
  $t\theta = t\theta + P_{\text{objectangle}}$;
return MC;
END

As seen from Fig. 4, from each of these two data sets, we selected four images for training and one image for testing; for the test images, the color and texture characteristics of the ground objects can be observed in the training images, but the directions, sizes, and neighbor combinations of the ground objects different from those in the training images. Based on these data, our experimental design has the following characteristics.

3) Different Images Are Adopted as the Training Data and Test Data: Because deep neural networks are usually very large, they can easily "remember" a large number of details of the training data. Using the same image for training and testing will lead to very high classification accuracy, but this high accuracy represents only the ability to fit the training data, not the real classification ability; therefore, we use images that are different from the training images to evaluate the classification ability more fairly.

4) Single Training Image is Used for a Clear Sample Generation Assessment and Accuracy Comparison: It is easy to find the differences between a single training image and a test image. In the above-mentioned two data sets, training and the test image are quite different. On the one hand, the use of a single training image allows the sample generation ability of CGAN-TSIM to be observed more clearly; on the other hand, this situation provides a good basis for measuring the ability of a model to recognize many spatial characteristics in the test data that do not appear in the training set.

5) Effect of Increasing the Number of Training Images Is Investigated: We gradually increase the number of training images used from 1 to 4 to compare the accuracy achieved with fewer to more training images; in this way, the stability of a method can be tested. Furthermore, we use 23 training images for a more extensive test of model performance.

B. CGAN-TSIM Implementation and Parameter Settings

We adopted Python 3.6 and TensorFlow to implement CGAN-TSIM and performed all experiments on a computer

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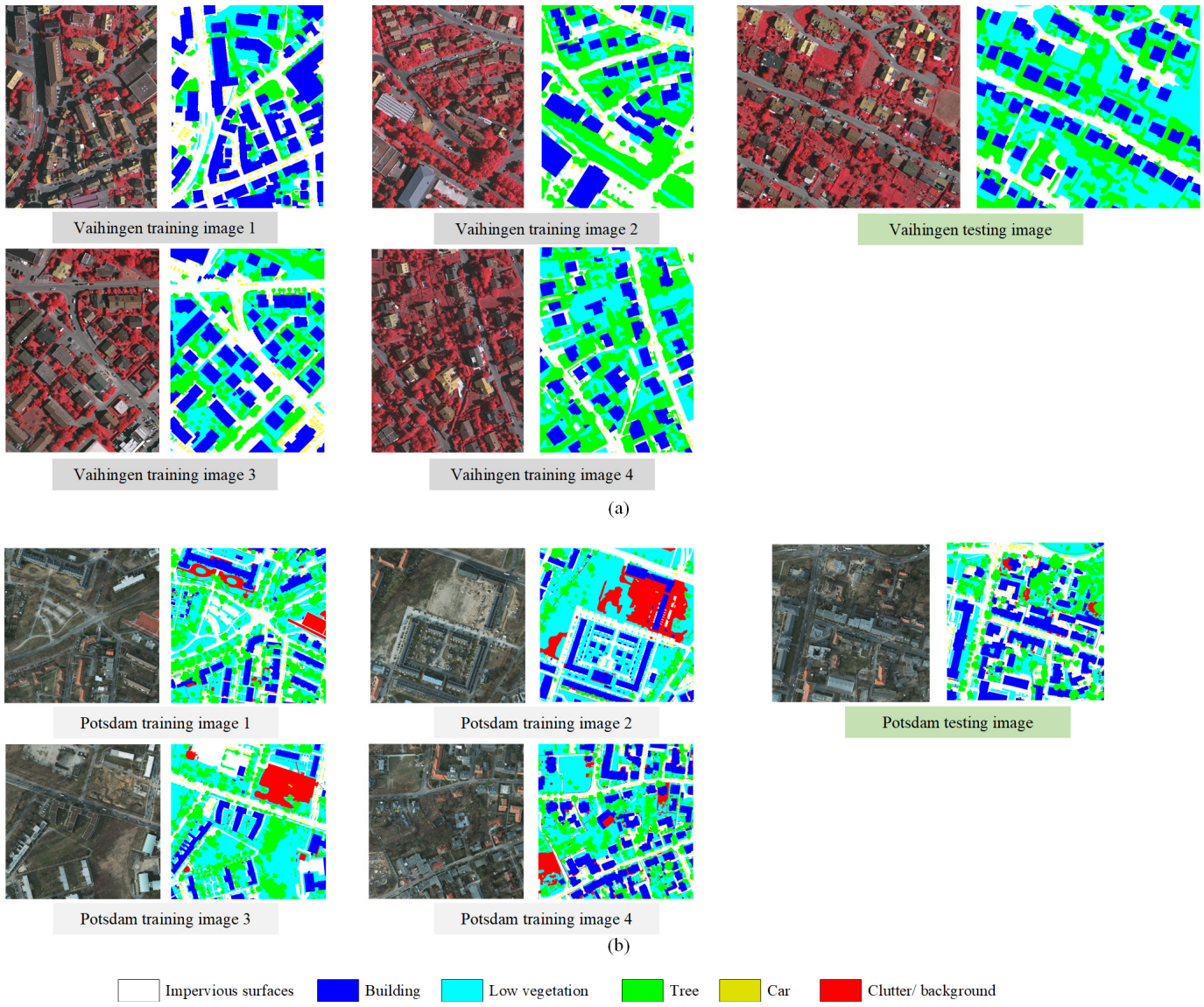


Fig. 4. Two test data sets. (a) Vaihingen data set. (b) Potsdam data set.

equipped with an Intel Core i9-9900K CPU, 64 GB of memory and an NVIDIA GeForce RTX 2080 Ti 11-GB GPU. All the input–output patch sizes for the submodels of CGAN-TSIM were set to 256 × 256. The CG model of CGAN-TSIM generated ground-truth images based on the two previously described strategies, the details of which are as follows.

1) Area separation strategy: The parameters $P_{\text{angleincrement}} = \pi/20$ and $P_{\text{offset}} = 256/10$ were adopted for all categories other than "cars." For each s_i in $S_{\text{sampleset}}$, the CG generated Permutation(4, 2) × 20 × 10 = 2400 ground-truth images for the Vaihingen data set and Permutation(5, 2) × 20 × 10 = 4000 ground-truth images for the Potsdam data set.
2) Object placement strategy for the "cars" category: In our training/test images, objects in the "cars" category always appear on "impervious surfaces." We set $P_{\text{objectangle}} = \pi/20$, $P_{\text{tileoffset}} = 128/10$, and a tile size of 128 × 128. For the Vaihingen data set, a car was defined as a 55 × 20 rectangle; for the Potsdam data set, a car was defined as a 100 × 36 rectangle. For each s_i in $S_{\text{sampleset}}$, this strategy resulted in 20 × 10 = 200 ground-truth images.

Using these two strategies, the CG generated ground-truth images to drive the CGAN to generate corresponding 256 × 256 remote sensing image patches to improve the training sample set.

C. Methods Considered for Comparison

The training and ground-truth images need to be cut into fixed-size image patches before they can be input into the neural network. In this study, four strategies were adopted to obtain the training image patches.

1) Default Cut Strategy: We sequentially separated each image into 256 × 256 patches from (0, 0) to the end of the image with a stride of (80, 80) pixels.

2) Finer Cut Strategy: We sequentially separated each image into 256 × 256 patches from (0, 0) to the end of the image with a stride of (40, 40) pixels. For a given image, the

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variation between the image patches obtained via the finer cut strategy (FCS) will be smaller than that between the patches obtained via the default cut strategy (DCS), but the number of patches will be $2 \times 2 = 4$ times as large.

3) Geometrical Transformation Strategy: Each image patch obtained via the DCS was subjected to additional geometrical transformations: rotation by 90°, 180°, and 270° and flipping along the horizontal and vertical axes. The number of patches obtained after the geometrical transformation strategy (GTS) was, thus, six times the original number obtained via the DCS.

4) CGAN-TSIM: We used the DCS to obtain the original image patches and CGAN-TSIM to improve the training sample set.

To analyze the impact of different training samples on a semantic segmentation neural network, we adopted the ordinary U-Net model as the semantic segmentation model; in this model, VGG-16 is used as the base pretrained encoder. The input–output size was set to 256×256. Based on U-Net and the four patch strategies, we compared five methods in our experiments.

1) U-Net + DCS:
   In this method, only the image patches obtained via the DCS were used as the training sample set.
2) U-Net + FCS:
   In this method, only the image patches obtained via the FCS were used as the training sample set.
3) U-Net + GTS:
   In this method, only the image patches obtained via the GTS were used as the training sample set.
4) U-Net + Adversarial Training:
   To compare our proposed method with a similar kind of method, we selected a previously proposed method in which an AT mechanism is introduced to improve the results of a semantic segmentation model [51]. In this method, the output of a DSSNN model is passed to a GAN to evaluate the higher order inconsistencies between the DSSNN output and the ground-truth segmentation maps; experimental results based on a computer vision test set show that this method can improve the semantic segmentation accuracy of the DSSNN.
5) U-Net + CGAN-TSIM:
   In this method, the improved training sample set obtained by applying CGAN-TSIM to the image patches obtained via the DCS was used.

D. Experimental Results and Analysis

1) Condition Generation and Image Generation: In the CGAN-TSIM training process, each training image is separately used to train the CGAN, giving the CGAN the ability to generate image patches based on the characteristics of the input image. The format of the images generated by the CGAN is consistent with the format of the input sample. For the Vaihingen data set, each generated image patch contains three bands (near-infrared, red, and green), the size is 256×256, and the resolution is 9 cm. For the Potsdam data set, each generated image patch contains three bands (near-infrared, red, and green), the size is 256×256, and the resolution is 5 cm. For the two data sets, examples of the generated image patches, real-image patches and real-ground-truth image patches obtained based on one training image from each data set (Vaihingen training and Potsdam training) are shown in Fig. 5.

As shown in Fig. 5(a) and (b), for these training image patches, each image patch generated by the CGAN is extremely similar to the corresponding real image patch; except for some very small details and lines, the data generated by the CGAN are hardly distinguishable from the real data. The CG was then used to generate additional ground-truth images based on the two previously defined strategies, and these new ground-truth images were processed by the CGAN to generate corresponding remote sensing image samples. Examples of the created ground-truth images and the corresponding generated image samples are shown in Fig. 6.

As shown in Fig. 6(a), based on either ground-truth images generated by the CG or ground-truth images provided by the user (the letter "A" image), the GAN can generate corresponding remote sensing images. For the same ground-truth image, the images generated by the CGAN for the Vaihingen and Potsdam data sets differ greatly. During the generation process, the CGAN does not simply create images with specific pixel colors and stitch them together; instead, the images also embody certain spatial relationships. As shown in Fig. 6(b), for the Vaihingen data set, we can observe shadows of higher objects cast on lower objects when the higher objects are located to the right of the lower objects, but such shadows do not exist when the locations of the higher and lower objects are swapped; this clearly reveals the directionality of the generated results. By contrast, there is no such obvious directionality characteristic in images generated based on the Potsdam data set; the main differences between trees and low vegetation lie in their textures. These differences are evident in Fig. 6(a), where the generated letter "A" images consist of impervious surfaces surrounded by trees. The generated image corresponding to the Vaihingen data set contains directional shadows, and consequently, the "A" has a significant visual effect indicative of depression, while the generated image corresponding to the Potsdam data set has no such effect.

2) Analysis of Condition Generation and Image Generation Results: As a deep model, the CGAN of CGAN-TSIM contains numerous neurons. Since the number of input image patches is limited, this large model is easy to fit the available training image patches. Therefore, as shown in Fig. 5(a) and (b), when the ground truths for the original training images are provided as input, the CGAN can generate fake images that are highly similar to the real images (although the generated images appear slightly blurred). Images generated in this way from the training ground truths can be regarded as merely a low-quality repetition of the original training data; this repetition is not helpful in capturing more diverse boundaries, directions, and combinations of ground objects. Consequently, these data generated from the original ground truths of the training images cannot enhance the training data set.

During the CGAN training process, the characteristic spatial information contained in the training image is captured by the deep neural network; then, based on new ground-truth

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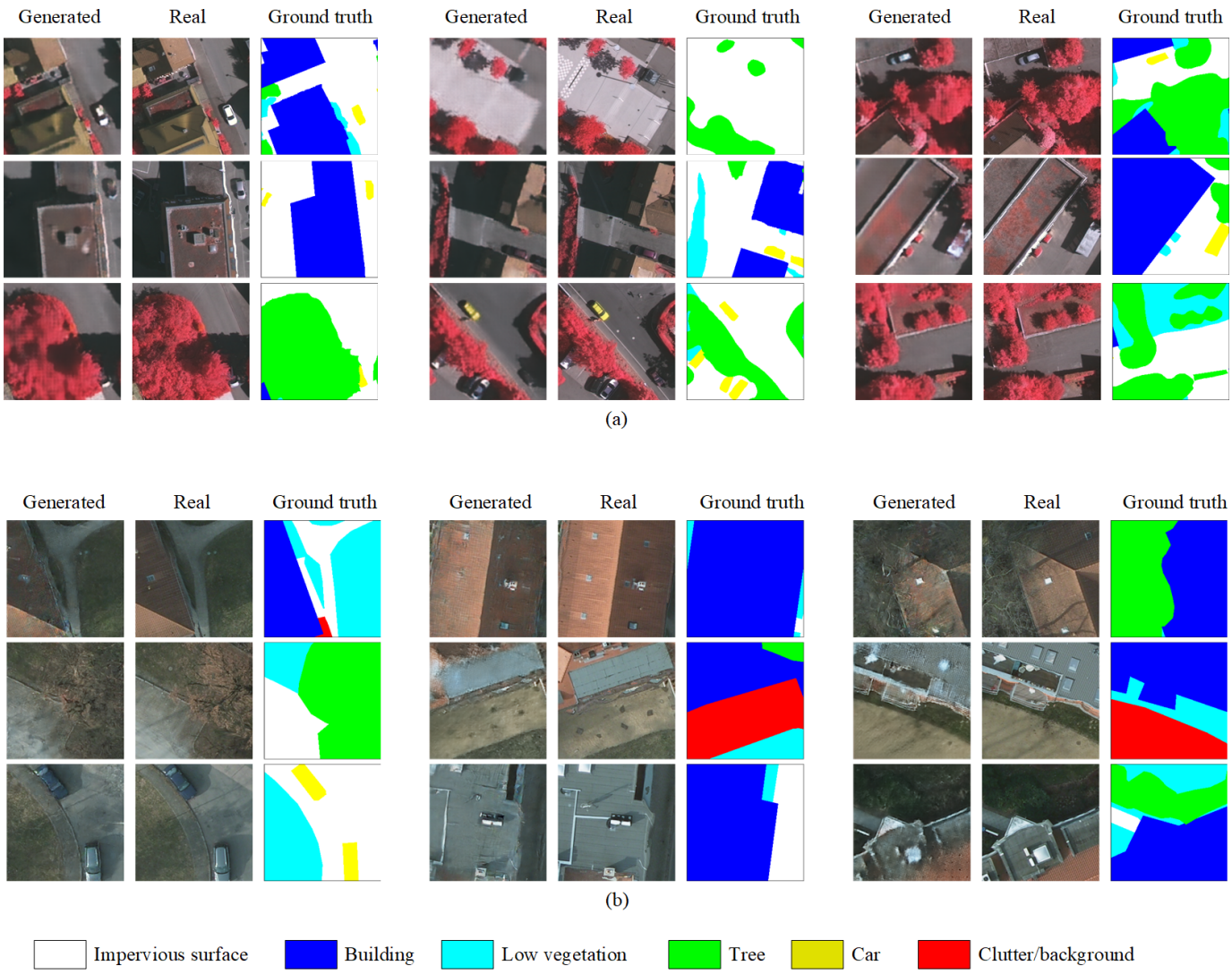


Fig. 5. Generated image patches based on the ground truths for real training image patches. (a) Generated results based on Vaihingen training. (b) Generated results based on Potsdam training.

images generated by the CG, this captured knowledge can be embodied in corresponding images generated by the CGAN. Although the generated images in Fig. 6(a) and (b) are far rougher than those in Fig. 5(a) and (b), it can also be seen that this generation process does not constitute merely a simple pixel-to-pixel conversion; instead, realistic spatial relationships between adjacent objects, especially spatial relationships at boundaries, are reproduced. The Vaihingen data set contains obvious directional shadows, and therefore, higher elevation objects (buildings or trees) in certain orientations or combinations cast shadows on lower elevation ground objects (impervious surfaces or low vegetation). Cars can also cast shadows of similar shapes on impervious surfaces. By contrast, the shadows in the Potsdam images are insignificant; therefore, there are also no obvious shadows in the generated images, but we can see distinct differences in texture between objects belonging to similar categories, which are also a manifestation of the spatial relationships captured from the original data. These characteristics allow the generated images to fully reproduce the spatial relationships observed in the training images, especially those at boundaries. Using the ground-truth images generated by the CG, we can obtain images that are similar to yet different from the original training data; they contain more diverse object boundaries, directions, and combinations. These augmented images can be used as samples to train a DSSNN to improve its classification accuracy at boundaries, thereby improving the overall segmentation results.

3) Comparison of Different Segmentation Methods Trained on One Training Image: In this experiment, we adopted one training image from each data set (Vaihingen training and Potsdam training) to train each of the five methods considered for comparison. After training, each method was used to perform semantic segmentation of the test images. The semantic segmentation results of the five methods for the Vaihingen test image are shown in Fig. 7.

Fig. 7(a) shows that because only one training image was used, the performance of U-Net is not sufficient.

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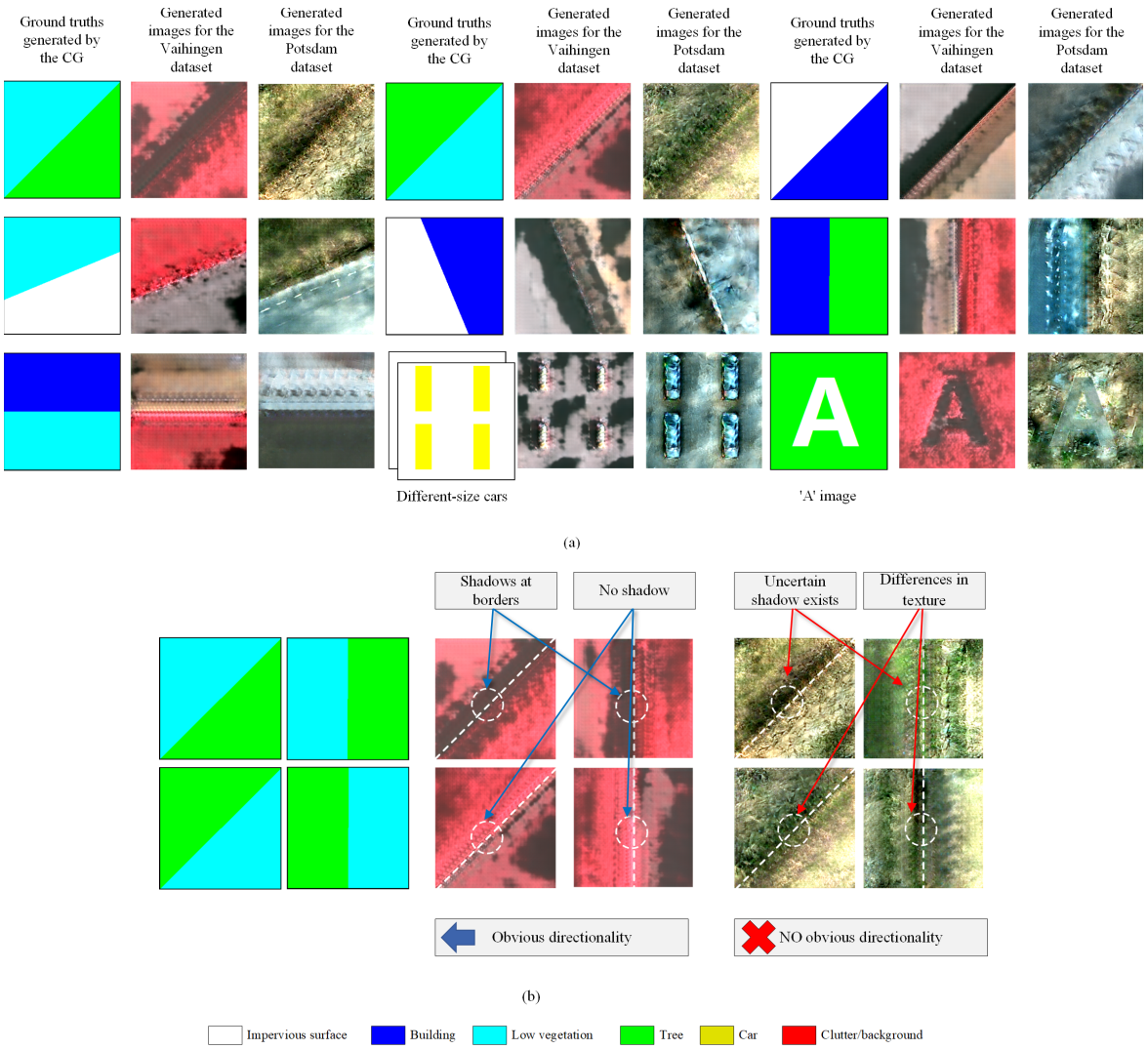


Fig. 6. Generated ground-truth images and corresponding generated image samples. (a) Generated images. (b) Generated results and spatial relations.

The U-Net + DCS, U-Net + FCS, and U-Net + AT results suffer from many errors at building boundaries, and many broken fragments exist in the results for the whole image. U-Net + GTS achieves better effects at building boundaries but misclassifies many areas with low vegetation and trees. We select three representative locations to show a detailed comparison of the segmentation results in Fig. 7(b). At location 1, in the first ellipse, the building boundaries in the U-Net + DCS, U-Net + FCS, and U-Net + AT results contain considerable amounts of deformation, and in the second ellipse, large areas of low vegetation are misclassified as trees; by contrast, the building boundaries in the U-Net + GTS results are quite close to the ground truth (even better than those of U-Net + CGAN-TSIM), but larger areas of low vegetation are misclassified. Meanwhile, U-Net + CGAN-TSIM classifies building boundaries well, and only small areas of low vegetation are misclassified; thus, overall, the classification results of U-Net + CGAN-TSIM are better than those of the other four methods. CGAN-TSIM also achieves the best results for location 2. Location 3 is a "**special**" location in the following sense: the area contained within the white ellipse in the image is a small section of road (impervious surface), and because of its small area, it does not appear in the ground-truth image (the ground truths in the Vaihingen data set always miss such small areas). However, all five methods successfully identify this area; in fact, they perform better than the ground truth in this area. Regarding the shapes in this area, CGAN-TSIM more faithfully reflects the shape of the road than do the other four methods.

Similar to Fig. 7(a), Fig. 8(a) reveals that there are many errors in the segmentation results for the Potsdam test image. Fig. 8(b) also displays three selected representative locations. At location 1, the roof centerline of a building is misclassified as an impervious surface by U-Net + DCS, U-Net + FCS, and U-Net + AT; by contrast, U-Net + GTS does not make this mistake, and U-Net + CGAN-TSIM obtains better results than all of the other four methods. Location 2 contains two areas highlighted with ellipses. The area indicated by the larger ellipse corresponds to a tree. For this area, U-Net + GTS obtains the best result, which is almost consistent with the

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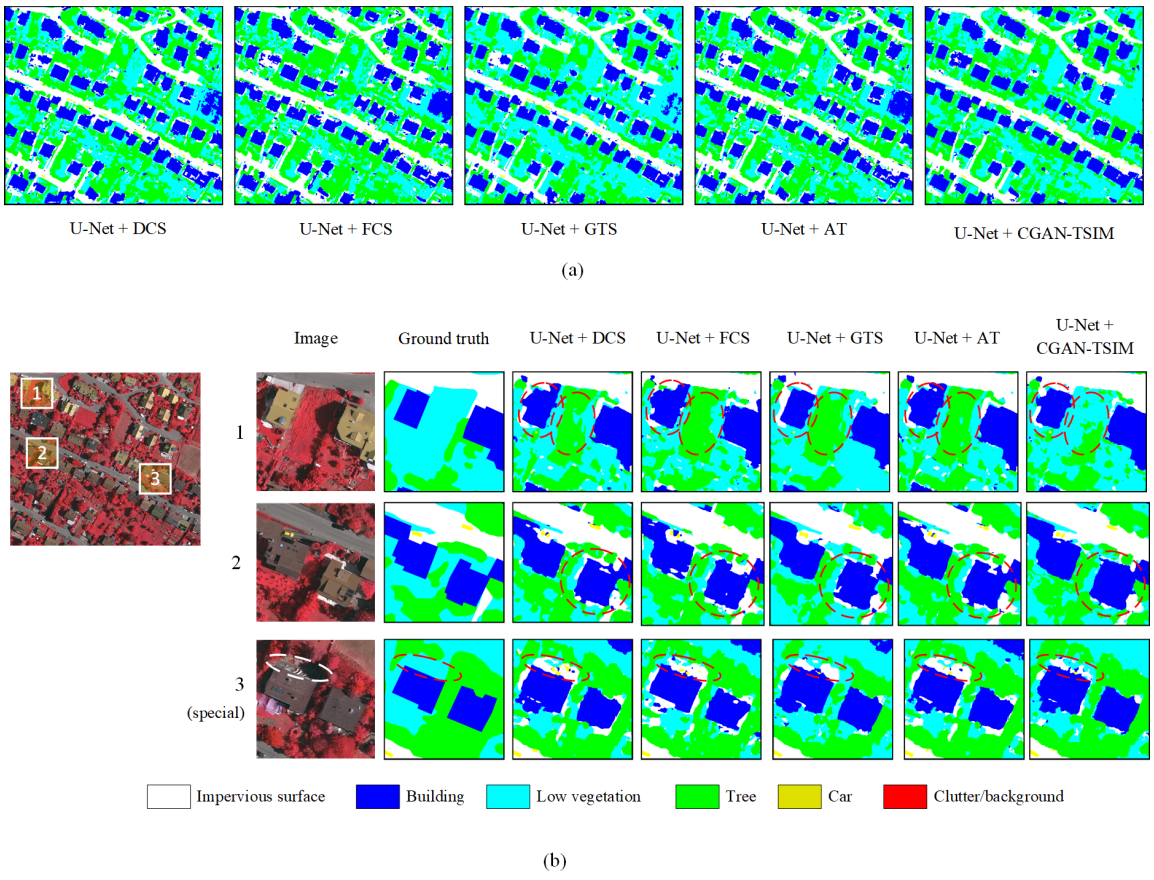


Fig. 7. Semantic segmentation results of the five methods for the Vaihingen test image. (a) Results for the whole test image. (b) Comparison of segmentation results at representative locations.

ground truth; unfortunately, however, in the area indicated by the smaller ellipse, the building recognition result of U-Net + GTS contains a large hole, which causes it to incorrectly classify the building as an impervious surface. By contrast, CGAN-TSIM obtains more balanced results than the other methods. Location 3 is a **special** location in the following sense: no features similar to the red marks (smaller ellipse) or multiple lines (larger ellipse) on the road appear anywhere in Potsdam training; the only red objects in the training image are roofs. Consequently, none of the methods is completely accurate in these areas, but the error of U-Net + CGAN-TSIM is the smallest.

Based on all the pixels of the ground truth image, we can obtain the overall classification accuracy of each method. The overall classification accuracies of the five methods for the two test images are listed in Table I.

As seen in Table I, for the Vaihingen test image, the accuracy of U-Net + DCS is only 72.12%. Compared with U-Net + DCS, U-Net + FCS and U-Net + AT show a slight decrease in accuracy, and the increase in the accuracy of U-Net + GTS is not obvious. For the Potsdam test image, the accuracies achieved by U-Net + FCS and U-Net + AT are slightly better than that of U-Net + DCS, and the improvement achieved with U-Net + GTS is more obvious than it is for the Vaihingen test image. U-Net + CGAN-TSIM achieves large improvements compared with the other four methods;

TABLE I
OVERALL CLASSIFICATION ACCURACIES OF THE FIVE METHODS FOR THE TWO TEST IMAGES

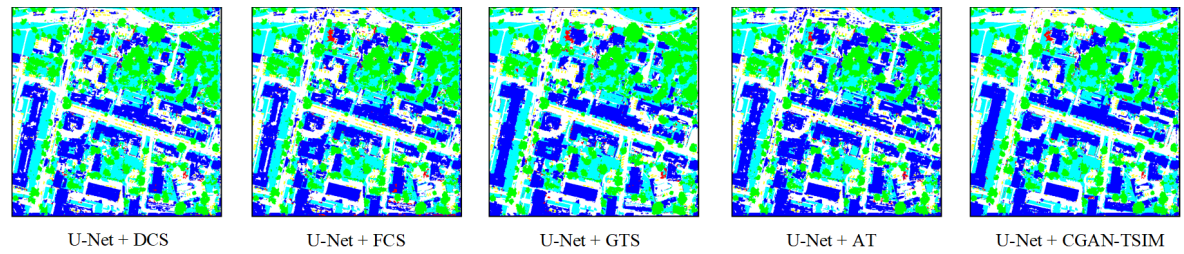
	Overall classification accuracy (%)	
Method	Vaihingen	Potsdam
U-Net + DCS	72.12	76.83
U-Net + FCS	71.61	77.12
U-NET + GTS	72.84	79.27
U-Net + AT	71.83	77.45
U-Net + CGAN-TSIM	**75.45**	**81.43**

its overall accuracy on the Vaihingen image is 75.45%, and on the Potsdam image, its overall accuracy is 81.43%.

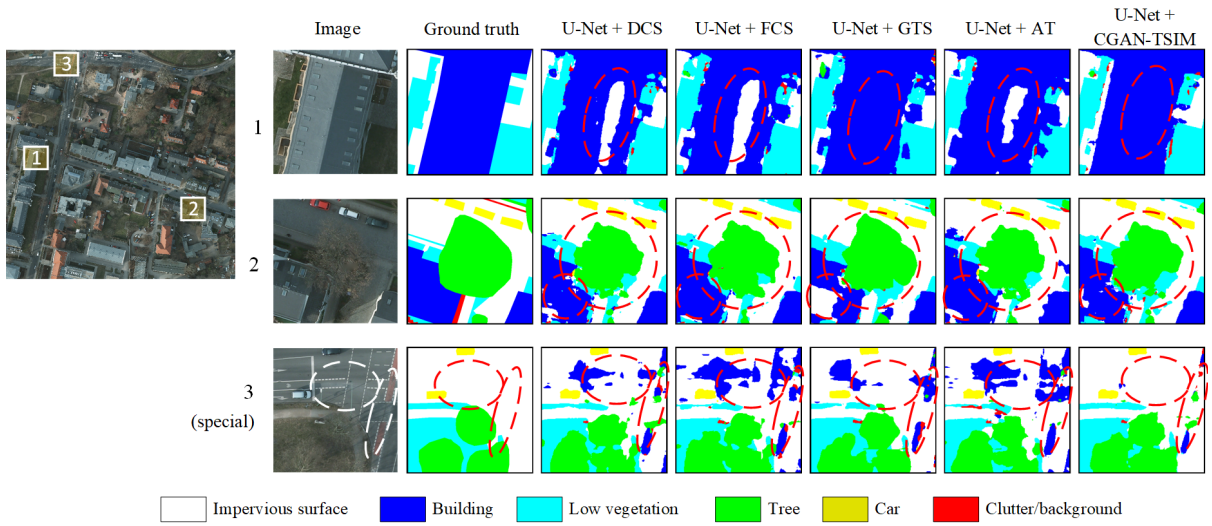
In addition, the per-category and average classification accuracies of the five methods for the two test images are listed in Table II.

As seen in Table II, for the Vaihingen data set, all five algorithms achieve their highest accuracy for buildings. Compared with the U-Net + DCS results, the accuracy results of U-Net + FCS and U-Net + AT are improved for cars but slightly decrease for the other categories; U-Net + GTS achieves the best accuracy of 94.25% for buildings, but its accuracy is lower for the categories of low vegetation and impervious surfaces. By contrast, CGAN-TSIM shows increased accuracy for all categories, and it achieves the best results among all five methods for the categories of low

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(a)



(b)

Fig. 8. Semantic segmentation results of the five methods for the Potsdam test image. (a) Results for the whole test image. (b) Comparison of segmentation results at representative locations.

TABLE II

PER-CATEGORY AND AVERAGE CLASSIFICATION ACCURACIES OF THE FIVE METHODS FOR THE TWO TEST IMAGES

Dataset	Method	Per-category accuracy (%)						Average accuracy (%)
		Low vegetation	Trees	Buildings	Impervious surfaces	Cars	Clutter/ background	
Vaihingen	U-Net + DCS	54.26	77.40	91.86	79.99	60.76	\	72.86
	U-Net + FCS	53.60	77.31	91.02	78.91	61.15	\	72.40
	U-NET + GTS	53.91	79.28	**94.25**	78.56	64.82	\	74.16
	U-Net + AT	54.06	77.09	91.32	79.31	61.80	\	72.71
	U-Net + CGAN-TSIM	**60.71**	**79.31**	93.03	**81.09**	**65.12**	\	**75.85**
Potsdam	U-Net + DCS	80.63	72.14	76.12	80.12	77.95	50.89	72.98
	U-Net + FCS	80.92	72.23	76.01	81.02	78.11	50.03	73.05
	U-NET + GTS	80.27	**81.43**	78.29	79.59	79.37	**58.11**	76.18
	U-Net + AT	81.43	73.41	76.15	80.71	78.27	51.11	73.51
	U-Net + CGAN-TSIM	**82.16**	81.06	**82.47**	**82.35**	**81.89**	57.69	**77.94**

vegetation, trees, impervious surfaces, and cars. For the Potsdam data set, if the U-Net + DCS results are again taken as the baseline for comparison, U-Net + FCS achieves improvements in four categories, but these improvements are very small. U-Net + GTS achieves an obvious improvement in accuracy for trees, buildings and clutter/background, but its accuracy is decreased for the low vegetation and impervious surfaces categories. With CGAN-TSIM, obvious improvements in accuracy are achieved. Since CGAN-TSIM shows improved accuracy for all categories, this method also achieves the best average accuracies: 75.85% for the Vaihingen data set and 77.94% for the Potsdam data set.

4) Analysis of Segmentation Results: The classification accuracy of a DSSNN is strongly affected by the amount of data used to train it. Only one remote sensing image used as training data cannot exhaustively represent all of the possible spatial diversity of the categories in its image patches. Although Vaihingen training and Potsdam training contain colors and textures for the majority of possible ground objects, the test image still contains unique object directions, sizes, and neighbor combinations. Therefore, the U-Net + DCS classification results are poor, especially at ground boundaries.

For U-Net + FCS, four times as many image sample patches were used for training as were used for U-Net + DCS.

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These additional patches allow the neural network to "remember" more details from the training image. Unfortunately, the network is also excessively fit these details and consequently does not achieve good results. At location 3 in Fig. 7(b), we can observe that the manually interpreted ground-truth image is not 100% accurate at the level of small areas or individual pixels. At location 3 in Fig. 8(b), red marks appear on the road in the test image; in the training image, this color appears only on roofs or in sports fields, never on the road. Accordingly, excessive sensitivity to details can cause overfitting problems; thus, the accuracy of U-Net + FCS is even lower than that of U-Net + DCS for the Vaihingen test image.

U-Net + GTS is helpful in identifying some objects: at some locations, it achieves the best segmentation results among all the methods for certain objects. However, as seen at location 1 in Fig. 7(b) and location 2 in Fig. 8(b), this advantage comes at a cost; while the results for some objects are better, the results for others may also be worse (even worse than those of U-Net + DCS). Consequently, U-Net + GTS achieves comparatively limited improvements in overall accuracy.

U-Net + AT adopts an AT mechanism to correct inconsistencies between the ground-truth patches and the segmentation results; this process makes the U-Net + AT results more similar to the input training image. For a small training data set, the DSSNN can easily achieve more than 95% accuracy on the training image. This strategy causes the neural network to remember more image characteristics; thus, U-Net + AT achieves a classification accuracy that is similar to (and slightly better than) that of U-Net + FCS.

U-Net + FCS, U-Net + GTS, and U-Net + AT are all still limited by the structural and directional characteristics of the objects in the original training image. In a single training sample image, many possible object directions or combinations will not appear, and these methods cannot directly resolve this problem. By contrast, U-Net + CGAN-TSIM adopts an active sample generation mechanism. Through the implemented condition generation strategies, more diverse category combinations and directional possibilities can be covered reasonably well in the image patches. When a DSSNN is trained on these data, the deep model will be able to handle more diverse classification situations; therefore, U-Net + CGAN-TSIM obtains the best results among all methods.

5) Comparison of Segmentation Results Obtained Based on Multiple Training Images: To further analyze the characteristics of the proposed method, we constructed four training groups for each of the two data sets using the following strategy. Group 1 contained only the first training image in the data set, Group 2 contained two images (1 and 2), Group 3 contained three images (1, 2, and 3), and finally, all training images were used in Group 4. For the four groups of training images for each of the two data sets, the overall classification accuracies of all methods are listed in Table III.

It can be seen from Table III that as more training images are added, the classification accuracy gradually improves for all five methods, indicating that adding more training images helps improve the classification accuracy for unknown images. The trends of variation in the classification accuracy are shown in Fig. 9.

Fig. 9(a) and (c) show that as the number of training images increases, the classification accuracy improves for all methods, but the improvement is not linear. As seen from Fig. 9(b), for the Vaihingen data set, the FCS, GTS, and AT methods cannot guarantee superior classification accuracy compared with the DCS results; in some cases, these strategies result in worse classification accuracy than the DCS. Fig. 9(d) shows that for the Potsdam data set, the FCS, GTS, and AT methods always yield better classification accuracies than the DCS. Among all methods, U-Net + CGAN-TSIM consistently achieves the highest classification accuracy and the greatest accuracy improvements compared with U-Net + DCS.

6) Analysis of Classification Results Based on Multiple Training Images: The above-mentioned results further demonstrate that a DSSNN model operates in a data-driven manner and that the use of more training data improves the model's classification ability. However, this does not mean that simply adding a certain number of image files will result in a proportional improvement; the different characteristics contained in different images can lead to uneven improvements in accuracy.

In semantic segmentation research, one common approach is to use strategies, such as the FCS and GTS to increase the number of sample image patches; however, how well these methods work is not specifically considered. From our experiments, we can see examples of undesirable results: for Vaihingen, the FCS and GTS do not result in obvious benefits, and in fact, they even cause decreases in accuracy in some cases. By contrast, the FCS and GTS always improve the accuracy of the Potsdam data set. The main reason for this difference may be that the Vaihingen images exhibit strong illumination and obvious directionality, while the Potsdam images do not possess such obvious characteristics. For Vaihingen, when the FCS is applied, the neural network may mistake certain directional shadows as ground object features and overfit them, leading to a decline in classification accuracy on the test image. Furthermore, the GTS will destroy the original directionality property of the training images, which can also lead to loss of the spatial relationships between shadows and objects.

For U-Net + AT, AT allows higher-order inconsistencies to be detected and corrected in the semantic segmentation network. Unfortunately, remote sensing images are usually quite large, and they need to be cut into patches for both training and classification. These image patches are usually small, and thus, may not contain an entire ground object; consequently, it is difficult for these patches to reflect the overall contours and shapes of such objects, and the correction of higher order inconsistencies does not work well. Therefore, using AT to improve the semantic segmentation accuracy for remote sensing images may not be a good choice.

Compared with the other methods, U-Net + CGAN-TSIM extracts object features at a higher level and actively constructs new samples with the goal of training the neural network. This strategy increases the number of sample

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TABLE III
CLASSIFICATION ACCURACY BASED ON EACH TRAINING GROUP FOR EACH OF THE TWO DATA SETS

	Overall classification accuracy (%)							
	Vaihingen training group				Potsdam training group			
Method	1	2	3	4	1	2	3	4
U-Net + DCS	72.12	73.77	77.31	78.84	76.83	77.02	79.31	82.53
U-Net + FCS	71.61	74.02	76.83	78.13	77.12	78.35	80.22	83.75
U-Net + GT	72.84	74.33	77.88	79.55	79.27	78.71	81.40	84.66
U-Net + AT	71.83	73.95	77.36	78.97	77.45	78.01	80.98	84.03
U-Net + CGAN-TSIM	**75.45**	**76.21**	**81.32**	**83.67**	**81.43**	**81.93**	**84.90**	**88.32**

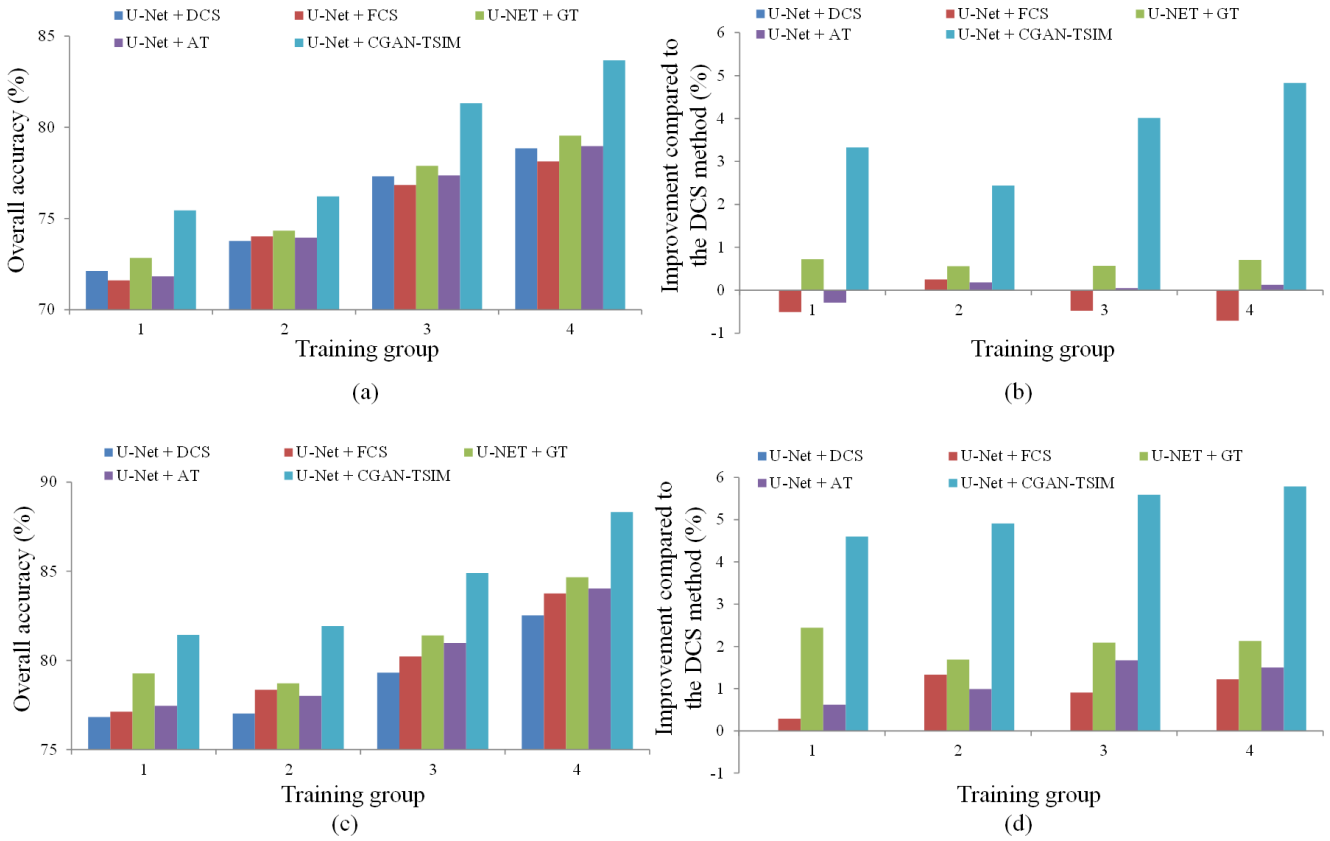


Fig. 9. Trends of classification accuracy variation. (a) Trends of variation in classification accuracy for Vaihingen. (b) Classification accuracy improvements of the other four methods compared with U-Net + DCS for Vaihingen. (c) Trends of variation in classification accuracy for Potsdam. (d) Classification accuracy improvements of the other four methods compared with U-Net + DCS for Potsdam.

images while preventing overfitting to specific characteristics; therefore, U-Net + CGAN-TSIM achieves greater accuracy improvements.

E. Experiment on a Larger Data Set and More DSSNN Models

The whole Vaihingen data set from "semantic labeling contest of the ISPRS WG III/4 data set" contains 33 semantically labeled images. To more extensively evaluate the capabilities of CGAN-TSIM, we selected ten images as test images and used the 23 remaining images as the training images. The ground objects in the ten test images possess various sizes, direction, color, and texture characteristics, and these images are listed in Fig. 10.

Moreover, additional DSSNNs, including FCN8s, FCN16s, SegNet, and U-Net, were tested on these images. For all images, the accuracies achieved with and without CGAN-TSIM are presented in Table IV.

Each row of Table IV reports the classification accuracies of all of the DSSNNs corresponding to a specific image, among which the highest accuracy is marked in bold; from the locations of these bolded results, we can identify that the performance achieved with CGAN-TSIM is significantly better than that without CGAN-TSIM. Thus, for all neural networks and test images, CGAN-TSIM improves the classification accuracy. The last row of Table IV presents the average accuracy improvement for each DSSNN when CGAN-TSIM is used; from this row, it can be seen that for all of the DSSNNs, the accuracy is improved by about 5% through the adoption of CGAN-TSIM, with U-Net achieving the maximum improvement of 5.42%. The above-mentioned results show that even in the case of a large amount of data, and for different

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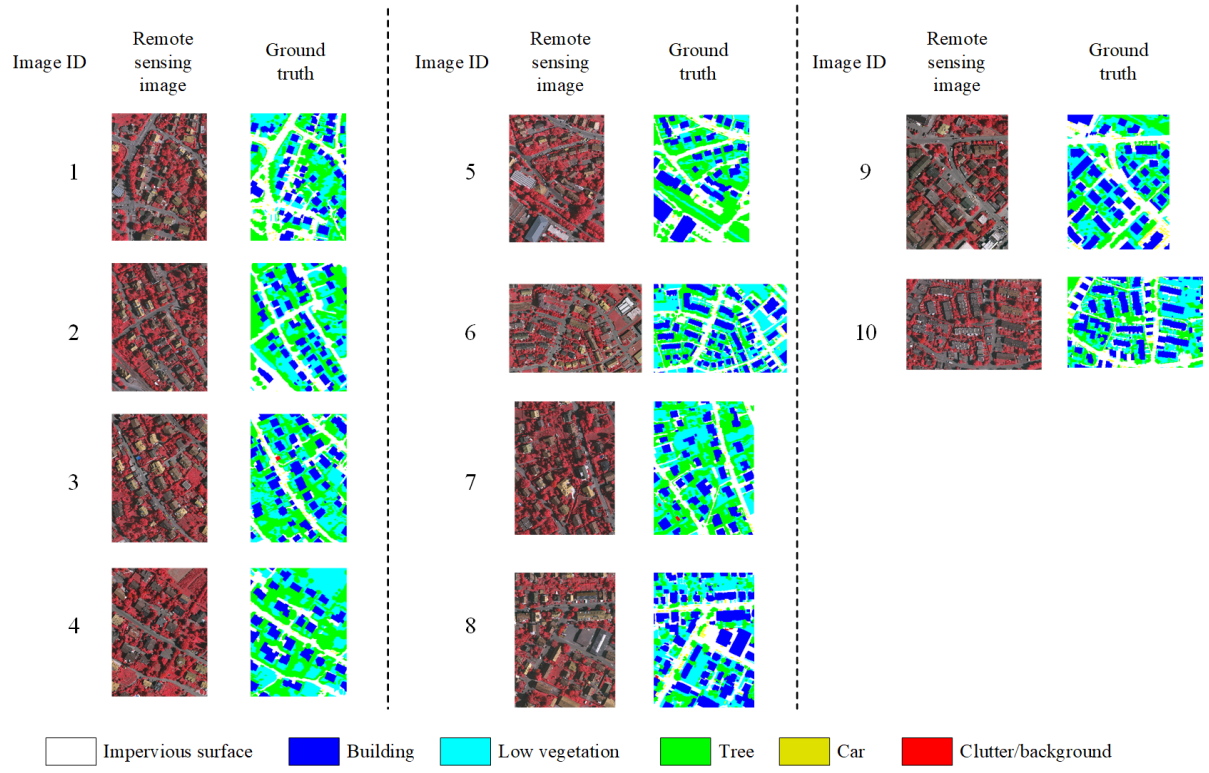


Fig. 10. Ten test images.

TABLE IV
OVERALL ACCURACY ON 20 IMAGES WITH AND WITHOUT CGAN-TSIM

Test Image	Without CGAN-TSIM (%)				With CGAN-TSIM (%)			
	FCN8s	FCN16s	SegNet	U-Net	FCN8s	FCN16s	SegNet	U-Net
1	84.90	82.11	85.03	85.78	89.88	86.69	90.09	**90.91**
2	81.89	80.35	81.07	81.91	**86.97**	84.91	86.00	86.89
3	81.64	80.85	82.73	82.64	86.56	85.58	88.04	**88.22**
4	83.73	82.37	84.05	84.04	88.66	87.06	88.96	**89.85**
5	83.80	83.93	85.49	86.58	88.86	88.48	90.68	**91.75**
6	85.49	82.21	84.33	86.32	90.41	87.08	89.46	**92.27**
7	81.44	80.68	83.12	82.08	86.16	85.03	**88.13**	87.46
8	80.67	81.33	84.17	83.22	85.79	86.24	**89.44**	88.25
9	85.70	82.15	85.33	86.95	90.62	86.61	90.61	**92.32**
10	80.15	82.27	83.73	84.45	85.30	87.39	89.09	**90.24**
Average accuracy improvement when using CGAN-TSIM					**4.98**	**4.68**	**5.15**	**5.42**

convolution/deconvolution operations and DSSNN models, the samples generated by CGAN-TSIM can still help to improve the DSSNNs' accuracy.

IV. CONCLUSION

The simplest and most direct way to improve the classification ability of a DSSNN model is to increase the size and quality of the training set. Providing an exhaustive set of category and neighborhood area variations for each pixel in the input–output image patches can endow a DSSNN with better inference capabilities when segmenting a new image. To achieve this goal, most semantic segmentation data sets used in computer vision research are extremely large. However, because the possible fields of application vary widely, a few images are available for many situations of interest. Moreover, due to funding/time limitations, it is difficult to construct large remote sensing data sets for real segmentation applications; therefore, it is necessary to increase the number of samples using automated methods.

With the DCS for obtaining input image patches as the baseline, two traditional methods (the FCS and the geometrical transformation strategy) of increasing the sample size were tested. The experimental results show that these two methods result in limited improvements in segmentation ability, primarily because these methods are still limited by the contents of the original training images themselves: they cannot create new content combinations that do not appear in the original images, and too many additional passively generated samples of this kind will result in overfitting and may even reduce the classification accuracy. AT is a method for training GANs that can be used to correct higher order inconsistencies in segmentation results. However, AT still enables only limited

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improvements in remote sensing segmentation, and overall, its improvements are not obvious.

Many of the possible object directions and category combinations will not be present in the original remote sensing images, which hinder traditional methods from producing training sample sets with sufficient spatial diversity. Therefore, rather than simply relying on finding samples from the original remote sensing images, the CGAN-TSIM proposed in this article adopts a new sample acquisition structure: first, a CG is constructed, which can automatically generate new ground-truth image patches for sample diversity improvement; second, these generated ground-truth image patches are used to drive a CGAN to generate corresponding remote sensing images; and finally, the generated ground-truth images and the corresponding generated remote sensing images are added to the original training sample set to improve the segmentation ability of a DSSNN. The novel features of CGAN-TSIM are listed as follows.

1) *Autonomous Sample Generation:* The CG can adopt a variety of strategies for creating ground-truth images. The objective of sample creation can be represented in the form of strategies or strategy parameters. This endows the CGAN-TSIM method with the important characteristic of autonomy. Unlike in traditional sample set enlargement methods, which passively find samples from images, users of CGAN-TSIM can actively decide the number and contents of the samples to be generated. This feature allows us to avoid simply aimlessly increasing the number of samples used in the training stage and rather to focus on what is truly needed to increase the spatial diversity of the samples.
2) *High-Level Spatial Information Representation:* The CGAN can extract high-level spatial information on objects in remote sensing images. The new samples are not generated merely through color and texture stitching but rather represent realistic spatial information and object relationships. This allows the sample generation objective to be translated into image patches with a high-level representation of spatial information.
3) *Stable Accuracy Improvement:* CGAN-TSIM generates sample patches that reflect more diverse object combinations and boundaries and do not contain special, rare or incorrect details from the training images; thus, CGAN-TSIM enables a relatively stable improvement in classification accuracy while avoiding overfitting.

CGAN-TSIM achieves better results than other methods because it enhances the spatial diversity of sample image patches, and thus, prevents overfitting to the details of the training images. The experimental results demonstrate that CGAN-TSIM leads to improved segmentation results and overall classification accuracy in comparison with four traditional methods. Hence, CGAN-TSIM is expected to play an important role in constructing DSSNN models with better classification capabilities for remote sensing images.

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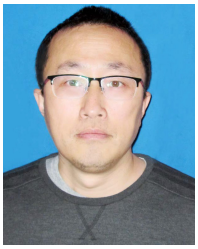
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